SafelightIO

USER MANUAL · FIRMWARE 1.0

The darkroom meter, explained

From the first test strip to a finished print.

THE TONE SCALE



Inside

Five tools: Exposure, Contrast, Split Grade, Virtual Proof, Control Panel.

Every control and readout, with screen maps and step diagrams.

Paper and sensor calibration, saving, backups and troubleshooting.

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> **Idea**
> The meter measures the light on your easel and turns it into a printing time: **light × your paper's constant = seconds**. Every tool in the device is built on that one number. Once your paper is calibrated, any new light reading becomes a time immediately.

How to use this manual

There are three ways in:

If you want to…	Start at
Set the meter up and make a first print	**1. Start here**
Look up one control you can see on screen	**15. Reference → Index of controls**
Fix a problem	**13. Messages, Help Center and troubleshooting**

The manual follows the device: first the screen, then the five tabs, then the sidebar, then the jobs that need several steps (calibration, saving, updates).

Words in this manual. A **control** is something you tap, drag or type into. A **readout** is a number or message the device shows you. Names printed in **bold** match the labels on screen. The device is a small screen in a darkroom, so labels are often abbreviated; the full name is given the first time.

> **Note**
> The **Help** button at the bottom of the sidebar opens the **Help Center**: seven short tabs (Start, Settings, Exposure, Contrast, Split Grade, Virtual Proof and Paper Editor), a QR code to the project page, and the support address shown on the Start tab (**support@cyanowood.com**). Use it when you are standing at the enlarger and need a reminder.
>
> The Help Center is a quick reminder. This manual is the complete, current reference. If a Help page and this manual disagree, trust this manual.

> **Tip**
> The HTML edition (open `MANUAL.html` in a browser) has a **Darkroom** button that switches to red-on-black. Use it on a phone or tablet beside the enlarger.



Prototype unit. The screen in this photo shows an early build; the screen maps in this manual reflect firmware 1.0.

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Fifteen chapters, in the order you will use them.

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1. Start here

1.1 What the meter does

SafelightIO is a light meter for black-and-white enlarger printing. You put its sensor where the print will be, press **Measure LUX**, and it tells you how many seconds to expose the paper. It can also:

- recommend a **contrast filter grade** from two readings of the negative;
- work out a **split-grade** soft + hard exposure;
- paint a **preview grid** of the print from a series of readings;
- check that the enlarger's light is **even** across the easel.

It is not a densitometer and not a colour analyser. It measures the light that actually falls on the easel, which is what the paper responds to.

1.2 The one thing that makes it accurate

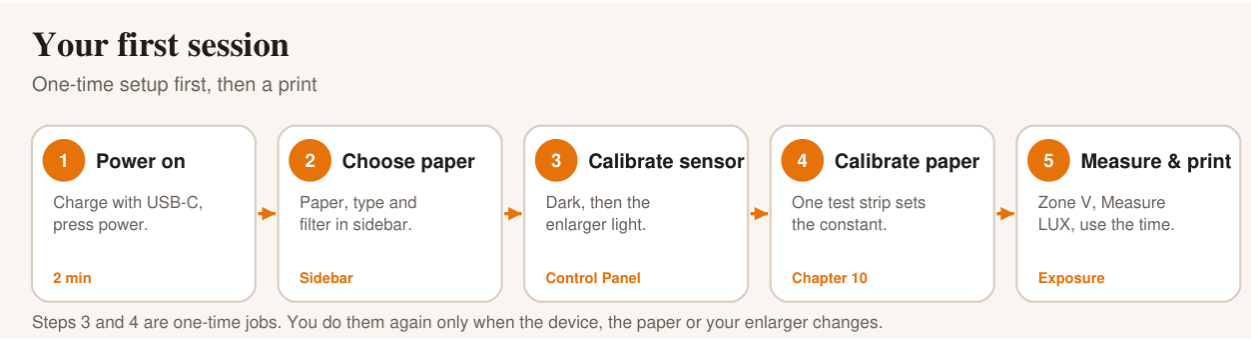
The meter needs one number from you: how sensitive **your** paper is under **your** enlarger. You find it once, with one test strip, and type the result into the device. That is **paper calibration** (chapter 10). After that, every reading becomes a time.

A meter that has not been calibrated for the selected paper cannot give a time. It will show `calibrate` instead. This is normal, not an error.

1.3 What you need

- The meter, charged or plugged in.
- Your enlarger, focused at the height and aperture you will print at.
- A sheet of the paper you intend to use, for the calibration test strip.
- A darkroom (or a very dark room) for the sensor's dark calibration.
- Optional: a microSD card. It keeps settings, papers and sessions on removable storage and makes backups easy.

1.4 Your first session



Step	What you do	Where	Time
1	Charge, power on, wait for the main screen	USB-C / power	2 min
2	Choose **Paper Brand**, **Paper Type** and **Filter Brand**	Sidebar	1 min
3	Calibrate the sensor: **Calibrate Dark**, then **Calibrate Light**	**Control Panel** tab	10 min
4	Calibrate the paper with one test strip	Chapter 10	15 min
5	Measure the easel and print	**Exposure** tab	1 min

Steps 3 and 4 are one-time jobs per device/paper. You do not repeat them for every print.

> **Important**
> The sensor calibration makes the lux numbers trustworthy. The paper calibration turns lux into seconds. Skipping the sensor step still gives a time, but it may be consistently too long or too short; skipping the paper step gives no time at all.

Step 1 — Power on

Connect USB-C to charge if the battery is low. Press the power button; the splash screen appears with a photography quote, a darkroom tip, the firmware banner (**SafelightIO Darkroom Meter v.1.0**, **Prototype - Not for Sale**) and the serial number. After a few seconds the main screen opens.

The backlight is deliberately dim while the device boots, then returns to your saved brightness. If the screen looks almost black, see **Brightness** in chapter 8 before assuming the meter is broken.

Step 2 — Choose the paper and filter

In the sidebar, set:

- **Filter Brand** — the brand of contrast filters in your enlarger (for example Ilford or FOMA). It changes the grade names and the built-in filter factors.
- **Paper Brand** and **Paper Type** — the paper in the easel.

Changing the paper reloads that paper's calibration and trims, and resets the dodge/burn slider. Choose the paper before you measure, not after.

Step 3 — Calibrate the sensor

Open the **Control Panel** tab and find **SENSOR CALIBRATION**:

1. **Calibrate Environment Noise** → **Calibrate Dark**, in total darkness.
2. **Calibrate Enlarger Light** → **Calibrate Light**, with the enlarger on and the sensor in steady light.

The full procedure, including the two-phase light calibration, is in section 7.4. Do this once per device; repeat it if readings disagree between gain settings or after a firmware update that changes the sensor settings.

Step 4 — Calibrate the paper

Follow chapter 10. You will make one test strip, choose the correctly exposed midtone, and enter its lux reading and time. The device computes the paper's constant. From then on, the **Exposure** tab gives times for that paper.

Step 5 — Measure and print

Open **Exposure**, choose the zone you measured (usually **V**), press **Measure LUX**, and read the time. Chapter 3 explains every control on that tab.

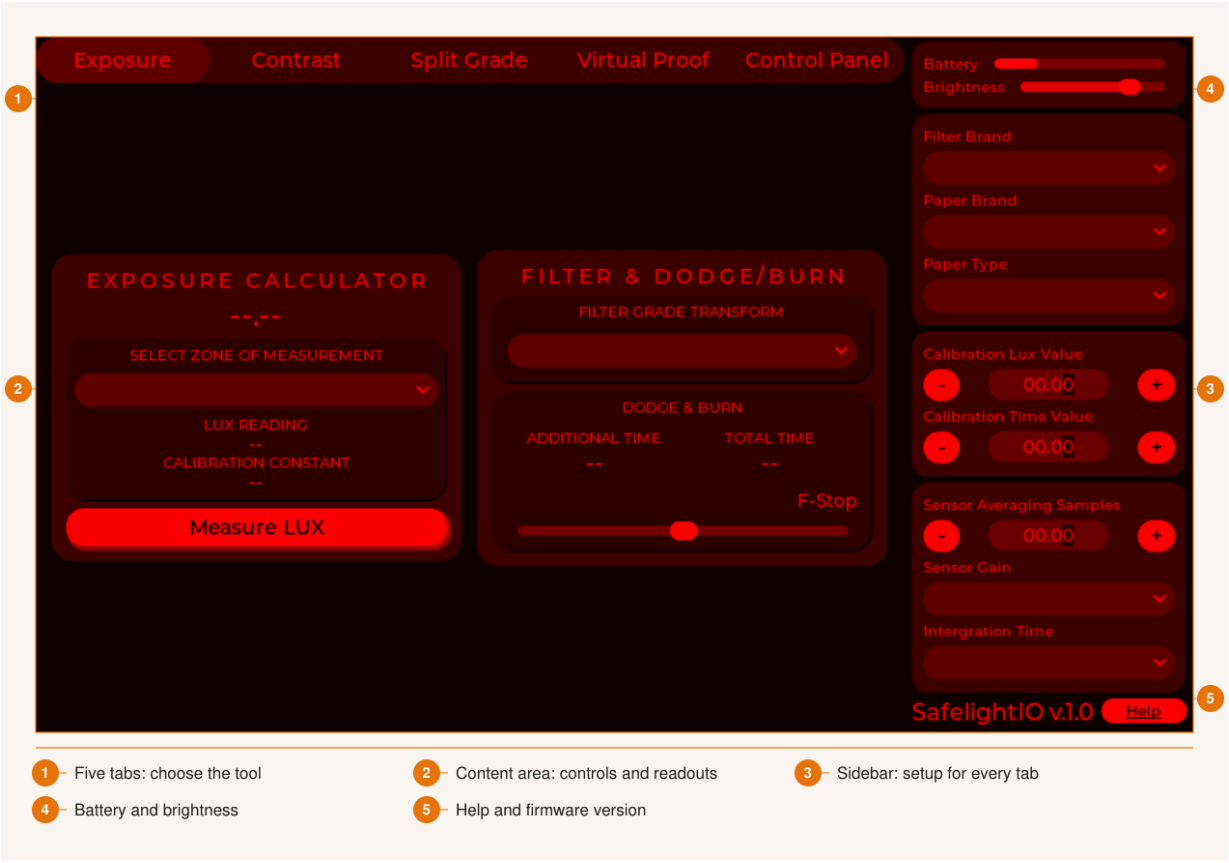
1.5 The five tools at a glance

Tab	Answers the question	Use it when
Exposure	"How long do I expose this print?"	You have a normal negative and want a time
Contrast	"Which grade does this negative need?"	The negative looks flat or contrasty
Split Grade	"What soft and hard exposures make this print?"	You want to control highlights and shadows separately
Virtual Proof	"What will the print look like across the easel?"	The negative is uneven, or you want to see before committing paper
Control Panel	"How is the meter set up?"	Filter data, paper estimates, saving, sensor calibration

The **Enlarger mode** switch lives inside Virtual Proof; it turns the same grid into a light-evenness test (section 6.5).

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2. The screen at a glance



The annotated screen above names the three parts. Each tool chapter opens with its own screenshot, so you can match the labels to the device.

The main screen has three parts:

- **Five tabs** across the top: Exposure, Contrast, Split Grade, Virtual Proof, Control Panel. The selected tab is highlighted.
- **The content area** in the middle: the controls and readouts for that tab.
- **The sidebar** on the right: paper, filter, sensor, calibration values, brightness, battery and the **Help** button. The sidebar is visible on every tab.

At the bottom of the sidebar the device shows **SafelightIO v.1.0**. This is the firmware version; it is also on the splash screen and in the Help Center.

> **Note**
> The device uses a dim red-on-black screen to protect your night vision. The screenshots in this manual reproduce that look; they can seem darker on paper than the screen does in a darkroom. They were taken before a reading, so values that have no measurement yet show `--`.

2.1 The tabs

Tab	What it holds	Chapter
Exposure	Zone, lux, calibration constant, exposure time, filter grade and dodge/burn	3
Contrast	Highlight and shadow readings, contrast range, grade, trims	4
Split Grade	Soft and hard exposures, filter pair, trims	5
Virtual Proof	Grid, histogram, selected sample, stability, enlarger mode	6

Control Panel	PAPER & FILTER EDITOR, ESTIMATE CALIBRATION, DATA MANAGER, SENSOR CALIBRATION	7

The fifth tab is labelled **Control Panel**. Its main heading is **PAPER & FILTER EDITOR**. The Help Center calls the same area **Paper Editor**. They are the same place; this manual uses **Control Panel** for the tab and **Paper & Filter Editor** for the editing section.

2.2 The sidebar

The sidebar is the meter's setup panel. It stays put while you change tabs. The full reference is chapter 8; in short:

- **Battery** and **Brightness** at the top.
- **Filter Brand**, **Paper Brand**, **Paper Type**.
- **Calibration Lux Value**, **Calibration Time Value**.
- **Sensor Averaging Samples**, **Sensor Gain**, **Intergration Time**.
- **SafelightIO v.1.0** and the **Help** button at the bottom.

> **Note**
> The correct spelling is Integration Time. The manual uses the correct spelling in prose and quotes **Intergration Time** when it refers to the on-screen label, so you can find it on the device.

2.3 Common actions

Action	What it does	Where
Measure LUX	Takes a light reading at the sensor	Exposure
Measure Highlight / **Measure Shadow**	Takes the two readings used for contrast	Contrast, Split Grade
Measure / **Measure next (n/total)**	Reads one proof cell and advances the grid cursor	Virtual Proof
Set V Zone / **Set Reference**	Uses the selected cell as the middle-gray anchor	Virtual Proof
Clear Grid	Erases the proof grid and its reference	Virtual Proof
Reset	Returns the current tab's trims to neutral (current paper only)	Contrast, Split Grade, Control Panel
Help	Opens the Help Center	Sidebar

2.4 Confirmation screens

Some actions cannot be undone, so the device asks first. The screen asks **Are you sure?** with **Yes** and **No**. It is used by:

- **Clear All Calibrations** (wipes every paper calibration and the reference);
- **Auto-Est. All Papers** (overwrites estimates for uncalibrated papers);
- **Load settings** and **Load Session** (replace current data);
- **Clear** under the sensor calibration headings;
- the second phase of **Calibrate Light**.

Read the question, then choose. **No** returns without changing anything.

2.5 The splash screen

At power-on, before the main screen:

- the banner **SafelightIO Darkroom Meter v.1.0**;
- **Prototype - Not for Sale**;
- an **Inspirational quote** and a practical darkroom tip (they change at random); for example **Tip: Pre-flash your paper for highlight detail**;

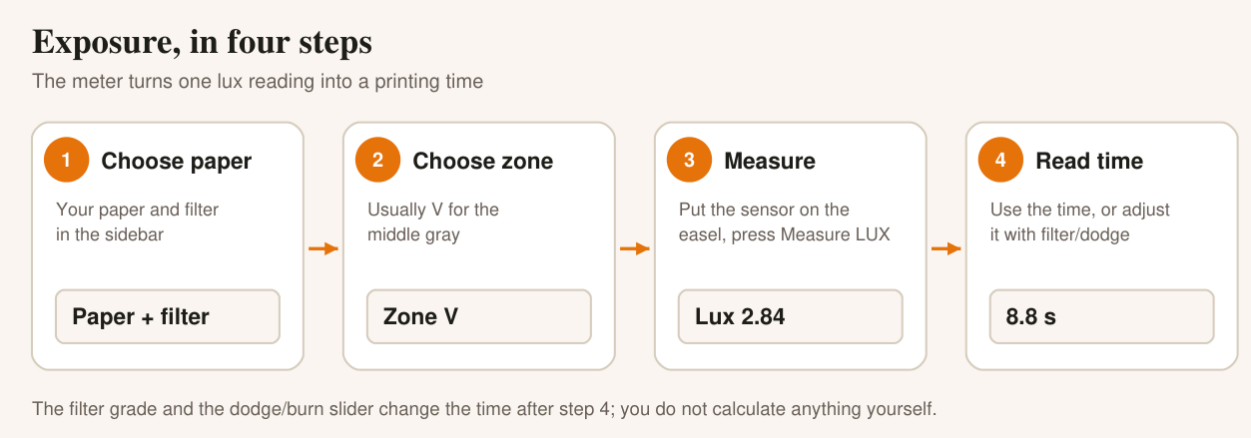
- a closing line such as **Have a happy developing sessions!**;
- the serial number and a QR code to the project page.

Nothing on the splash is a control. Wait for the main screen.

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3. Exposure

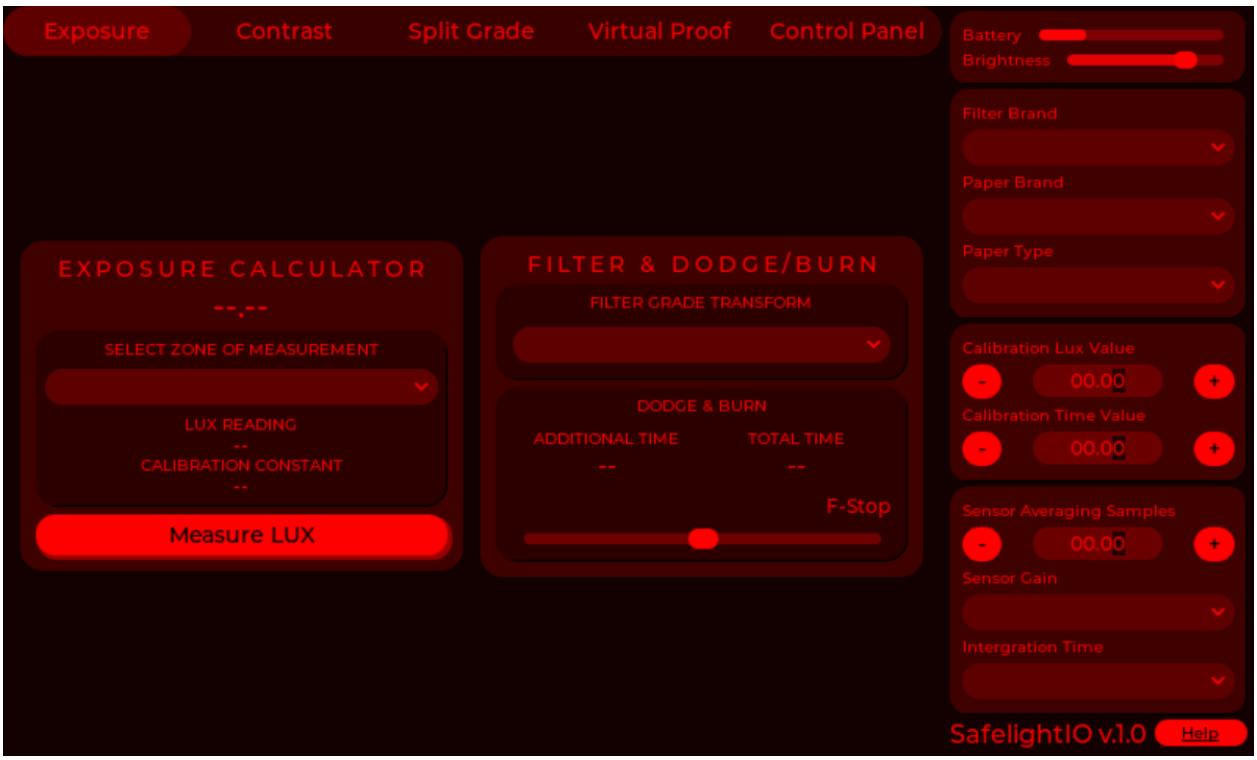
Use this tab when you have a negative and want a printing time.



3.1 The screen

The left column is the measurement: zone, lux reading, calibration constant and the **Measure LUX** button. The right column is the **FILTER & DODGE/BURN** panel: filter grade transform, dodge/burn slider, additional time and total time.

The large number at the top is the **exposure time**. Before you measure it may show a message instead of a number: `calibrate`, `measure lux`, `no sensor` or `error`. Chapter 13 explains each one.



3.2 Every control and readout

Element	What it is	What to do
EXPOSURE CALCULATOR	The tab heading	—
Exposure time readout	The suggested printing time in seconds	Read it; it updates after Measure and after filter/dodge changes

SELECT ZONE OF MEASUREMENT	Opens the zone list	Choose the zone you placed the sensor on; **V** is the normal midtone
Zone of measurement	The list: 0, I, II, III, IV, V, VI, VII, VIII, IX, X	See the zone table below
LUX READING	The last measured light level in lux	Shows the raw measurement
CALIBRATION CONSTANT	The paper's lux × seconds constant	Set by paper calibration; read-only here
Measure LUX	Takes a reading and calculates the time	Press once per measurement; the backlight dims while it reads
FILTER & DODGE/BURN	Heading for the right-hand panel	—
FILTER GRADE TRANSFORM	Opens the filter grade list	Choose the filter you will actually use. The grade's factor lengthens the time
DODGE & BURN	Heading for the slider and the two times	—
Dodge and burn slider	Adds or removes exposure in ¼-stop steps	Drag right to print darker, left to print lighter
ADDITIONAL TIME	The seconds added by the slider	Read-only; negative values mean less time
TOTAL TIME	Base time + additional time	This is the time to use at the enlarger
F-Stop	The slider's position expressed in f-stops	Read-only; useful when writing notes

3.3 Step by step

1. **Select the paper.** In the sidebar choose **Paper Brand** and **Paper Type**.
2. **Select the filter.** Choose the grade you will put in the enlarger under **FILTER GRADE TRANSFORM**. “No filter” keeps the time unchanged.
3. **Place the sensor.** Put it where the midtone will fall on the easel. Keep your body and the safelight out of the light path.
4. **Choose the zone. V** for a midtone; **III** for a shadow you want to place; **VII** for a highlight.
5. **Press Measure LUX.** Wait for the backlight to return.
6. **Read the time.** Use **TOTAL TIME** if you moved the dodge/burn slider; otherwise use the large exposure time.

3.4 The zone list

The zone tells the meter which tone you measured. A reading at **Zone V** is the reference. A reading at **Zone III** is two stops darker, so the meter multiplies the light to find the equivalent Zone V value.

Zone	Tone you are measuring	Meter factor
0	Deep black, almost no detail	×32
I	Black with a hint of detail	×16
II	Dark shadow with texture	×8
III	Shadow with good detail	×4
IV	Dark midtone	×2
V	Middle gray — the reference	×1
VI	Light midtone	÷2
VII	Highlight with texture	÷4
VIII	Bright highlight, little texture	÷8
IX	Near white	÷16

X	Paper white	÷32

> **Tip**
> Measure **V** when the sensor sits on the midtone. Measure **III** when you want to place a shadow, or **VII** for a highlight. The meter converts either reading to the same exposure scale, so you do not have to move the sensor to a midtone.

3.5 What the filter grade does

A contrast filter absorbs light, so it needs a longer exposure. The meter knows the built-in factor for each grade and brand, and multiplies the base time by that factor. The filter grade also changes the contrast of the print; the meter does not try to compensate for that, because that is the point of the filter.

If you have measured your own filter factor (chapter 12), the device uses your value instead of the built-in one.

3.6 Dodge and burn

The slider is a trim, not a second exposure. It adds or removes time from the whole print in ¼-stop steps, up to ±2.25 stops. Use it when the print is consistently a little light or dark, or when you want to try a variation without recalibrating.

- **ADDITIONAL TIME** shows the seconds the slider adds (or subtracts).
- **TOTAL TIME** is what you actually expose.
- The slider resets when you change paper.

For local dodging and burning under the enlarger, use the same total time as your starting point and cover/uncover areas as usual.

3.7 A worked example

A portrait negative, Ilford FP4, printed on Ilford Multigrade Classic Cooltone. The sensor is on the face at **Zone VI**.

1. The filter is grade **3**, factor about 2.5.
2. The sensor reads **2.84 lux** at Zone VI.
3. The meter converts Zone VI to the Zone V equivalent: 2.84 ÷ 2 = **1.42 lux**.
4. The paper's constant gives a base time of **8.80 s**.
5. The grade 3 factor makes it 8.80 × 2.5 ≈ **22 s**.
6. The dodge/burn slider at +2 quarter-stops adds about 10 s → **~32 s** total.

You do not have to do this arithmetic. The point is that every step has a visible readout, and the **TOTAL TIME** already includes all of them.

3.8 Common questions

Why does the time change when I change the filter? The filter absorbs light; the factor compensates. The print contrast changes too, which is why you chose the filter.

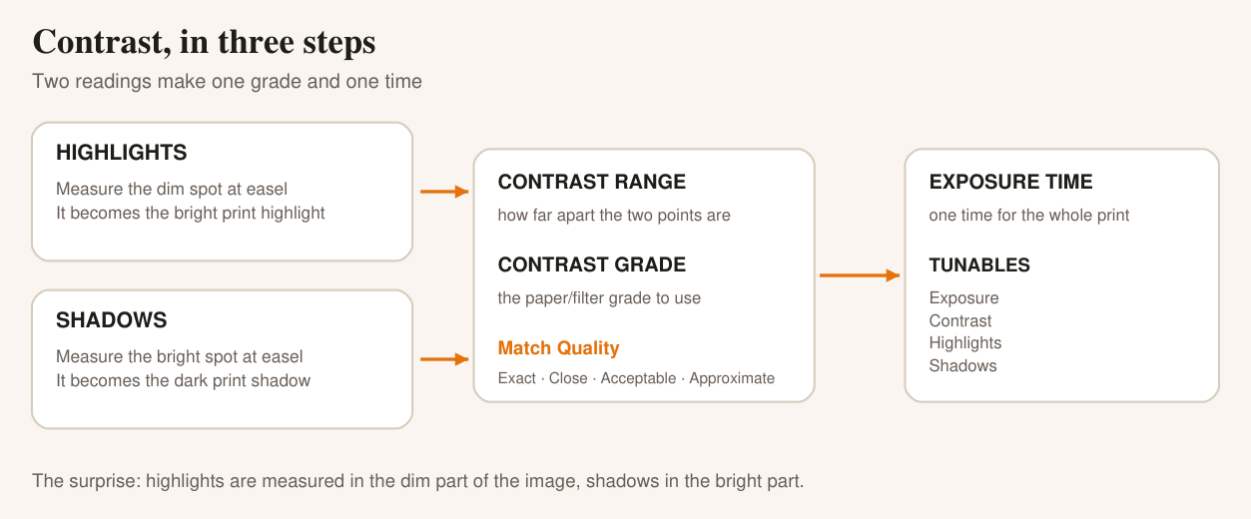
Why does the time change when I change paper? Each paper has its own calibration and its own trims.

Can I use the meter without calibrating the paper? You can read lux, but you will not get a time. The tab shows `calibrate` until the paper is calibrated.

The time looks wrong by exactly 2×. Check that you measured the zone you think you did, and that the enlarger height and aperture match the calibration. If they do, run **Calibrate Light** again.

4. Contrast

Use this tab when you want the meter to recommend a contrast grade.

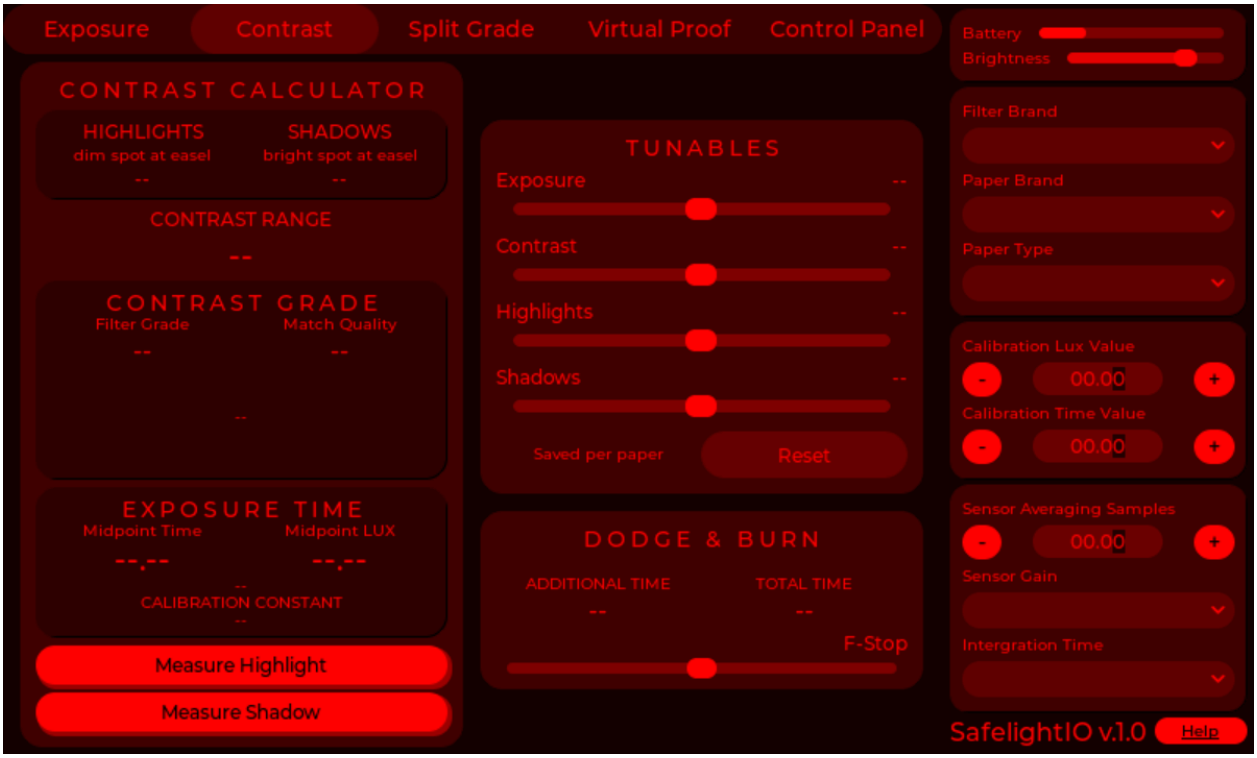


4.1 What the two readings mean

The meter needs two points in the projected image:

- **HIGHLIGHTS** — the **dim spot at easel**. A highlight on the print comes from a dense part of the negative, so less light reaches the easel.
- **SHADOWS** — the **bright spot at easel**. A shadow on the print comes from a clear part of the negative, so more light reaches the easel.

This is the part that surprises people. Measure the **dim** part for highlights and the **bright** part for shadows. The small labels under the headings say exactly that, so you do not have to remember.



4.2 Every control and readout

Element	What it is	What to do
CONTRAST CALCULATOR	The tab heading	—
HIGHLIGHTS + **dim spot at easel**	The highlight reading area	Put the sensor on the dimmest important highlight and press **Measure Highlight**
SHADOWS + **bright spot at easel**	The shadow reading area	Put the sensor on the brightest important shadow and press **Measure Shadow**
Highlight lux reading	The light measured at the highlight	Read-only
Shadow lux reading	The light measured at the shadow	Read-only
CONTRAST RANGE	The brightness range of the negative in stops	Read-only; roughly 3 stops is gentle, 5 stops is contrasty
CONTRAST GRADE	The grade the meter recommends	The starting point for your filter choice
Filter Grade	The grade actually selected for the calculation	Shows the recommended grade; used for the time
Match Quality	How closely the negative fits that grade: **Exact**, **Close**, **Acceptable** or **Approximate**	**Exact**/**Close** need no action; **Acceptable** is a compromise; **Approximate** means the range falls between grades
Analysis reasoning	The explanation line under Match Quality	Read why the device chose the grade
EXPOSURE TIME	The suggested time at the midpoint	Read it
Midpoint Time	The time for the midtone	Read-only; the time before trims
Midpoint LUX	The light level of the midtone, derived from the two readings	Read-only
Exposure Note	A note about the result (for example, when the range is extreme)	Read it; it often explains a warning
CALIBRATION CONSTANT	The paper's lux × seconds constant	Read-only
Measure Highlight	Takes the highlight reading	Press once, with the sensor on the dim spot
Measure Shadow	Takes the shadow reading	Press once, with the sensor on the bright spot
TUNABLES	The four trim sliders	Use only when you want to override the recommendation
Exposure	Shifts the whole exposure up or down, ±1.5 stops in 1/6-stop steps	Adjust if the print is consistently light/dark
Contrast	Biases the grade softer or harder, ±1.5 stops	Adjust if the print needs more/less contrast than recommended
Highlights	Trims the highlight end, ±1.5 stops	Adjust if highlights print too light or too dark
Shadows	Trims the shadow end, ±1.5 stops	Adjust if shadows print too light or too dark
Saved per paper	Tells you the trims belong to the selected paper	Remember: changing paper loads that paper's own trims
Reset	Returns the four trims to neutral for the current paper	Use after experimenting
DODGE & BURN	A final whole-print trim	See below
Additional time (Contrast)	The seconds the dodge/burn slider adds	Read-only
Total time (Contrast)	The time to expose	Use this at the enlarger
F-Stop	The dodge/burn slider position in stops	Read-only
Dodge and burn slider (Contrast)	Adds/removes exposure in ¼-stop steps, ±2.25 stops	Drag right to darken, left to lighten

4.3 Step by step

1. Select the paper in the sidebar.
2. Put the sensor on the **dimmest important highlight** and press **Measure Highlight**. Keep your body out of the light.
3. Put the sensor on the **brightest important shadow** and press **Measure Shadow**.
4. Read **CONTRAST RANGE**, **CONTRAST GRADE** and **Match Quality**.
5. Put the recommended grade in the enlarger. The **Filter Grade** readout already includes that grade's exposure factor.
6. Read **EXPOSURE TIME** (or **Total time** if you use the dodge/burn slider).
7. If the print is consistently off, use the **TUNABLES** and the **Reset** button.

> **Tip**
> Both readings are required. With one missing, the tab shows `--` and the reason is given in the note. Measure the missing point and the result appears immediately.

4.4 How the recommendation works, without the maths

The two readings give a **brightness range**. A negative with a small range is flat and needs a harder grade; a negative with a large range is contrasty and needs a softer grade. The meter compares the range with the paper's contrast data and picks the closest grade.

- **CONTRAST RANGE** is in stops of exposure.
- **CONTRAST GRADE** is the recommended filter grade.
- The **time** is taken from the middle of the two readings, so the midtone stays put when you change grade.
- **Match Quality** tells you how well the negative fits the paper's range.
- The **trims** let you move the result without recalibrating.

4.5 Fixed-grade papers

Some papers have one fixed contrast grade. For those papers the tab shows **Fixed grade** and:

- the trims are ignored;
- **Split Grade** is not applicable;
- in the **Control Panel**, the Factor and ISO R fields are disabled; only ISO P can be changed.

The eight fixed-grade papers are listed in chapter 11.2. If you are printing on one, use **Exposure** and choose the filter grade that matches the paper's grade.

4.6 A worked example

A landscape negative on Ilford Multigrade RC Deluxe:

- highlight reading **4.10 lux**;
- shadow reading **0.32 lux**;
- the range is about **3.7 stops**;
- the meter recommends grade **1** with **Close** match;
- **EXPOSURE TIME** is **14.2 s**.

A grade 1 filter is soft, which suits the contrasty negative. If the print's highlights are too bright, add a small **Highlights** trim; if the shadows are muddy, add a **Shadows** trim.

4.7 Common questions

The grade changes when I move one reading slightly. Small changes near a grade boundary can flip the recommendation. Look at **Match Quality**; **Close** means either neighbouring grade is reasonable.

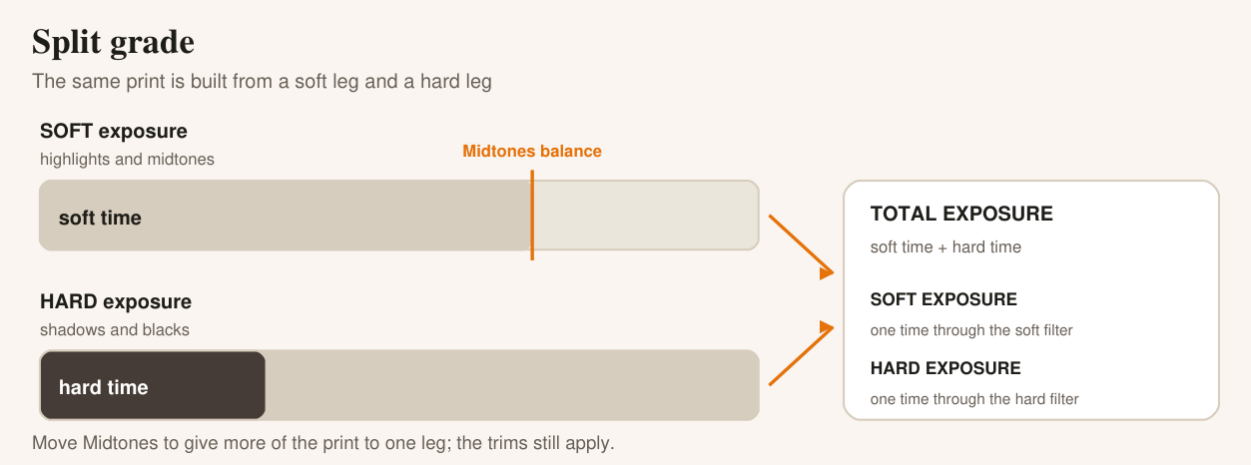
The time is not what I expected. The time is for the midpoint, not for the highlight. That is deliberate: it keeps the midtone stable when you change grade.

Should I use Contrast or Split Grade? Use Contrast when one grade suits the whole print. Use Split Grade when the shadows and highlights need different filters.

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5. Split Grade

Use this tab when shadows and highlights need different contrast.

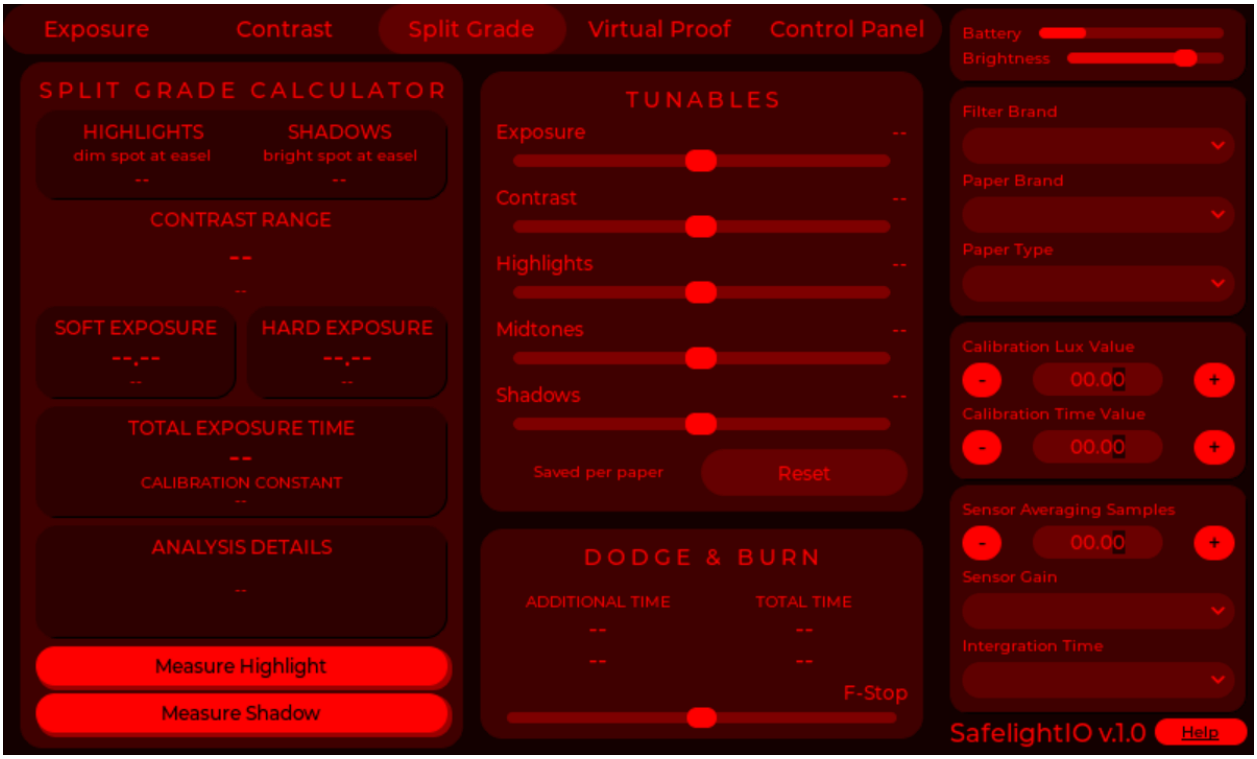


5.1 The idea

Split-grade printing gives the print two exposures through two filters:

- a **soft** filter for the highlights and midtones;
- a **hard** filter for the shadows and blacks.

The meter measures the same two points as the Contrast tab, then works out how much of the total exposure each leg should carry. The result is a pair of times that add up to the print you want.



5.2 Every control and readout

Element	What it is	What to do
SPLIT GRADE CALCULATOR	The tab heading	—

HIGHLIGHTS + dim spot at easel	Highlight measurement area	Measure the dim part of the image
SHADOWS + bright spot at easel	Shadow measurement area	Measure the bright part
Split highlight lux reading	Highlight light level	Read-only
Split shadow lux reading	Shadow light level	Read-only
CONTRAST RANGE	The negative's range in stops	Read-only
Equivalent grade readout	The single grade that would be closest to this split	Useful as a sanity check
SOFT EXPOSURE	The soft leg's filter name and time	Use this filter for the soft exposure
Soft filter name	The soft filter chosen by the meter	Changes with **Midtones** and paper
Soft exposure time	Seconds for the soft leg	Read-only
HARD EXPOSURE	The hard leg's filter name and time	Use this filter for the hard exposure
Hard filter name	The hard filter chosen by the meter	Changes with **Midtones** and paper
Hard exposure time	Seconds for the hard leg	Read-only
TOTAL EXPOSURE TIME	Soft + hard time	The total time the paper receives
CALIBRATION CONSTANT	The paper's constant	Read-only
ANALYSIS DETAILS	The reasoning: percentages of soft/hard light, and any warnings	Read it before printing
Measure Highlight / Measure Shadow	Take the two readings	Same as Contrast
TUNABLES	Five trims	Use after the first test print
Exposure	Shifts the total exposure, ±1.5 stops in 1/6-stop steps	Lighten/darken the whole print
Contrast	Biases the overall contrast, ±1.5 stops	More/less separation
Highlights	Trims the soft leg, ±1.5 stops	Adjust highlight density
Midtones	Chooses the split pair, ±1.5 stops	Moves the balance between soft and hard filters
Shadows	Trims the hard leg, ±1.5 stops	Adjust shadow density
Saved per paper	The trims belong to the selected paper	Changing paper loads its own trims
Reset	Returns all five trims to neutral	Use after experimenting
DODGE & BURN	Adds a final whole-print trim to both legs	See below
Additional soft time	Seconds added to the soft leg	Read-only
Total soft time	Soft leg after the trim	Use this at the enlarger
Additional hard time	Seconds added to the hard leg	Read-only
Total hard time	Hard leg after the trim	Use this at the enlarger
F-Stop	Dodge/burn slider position	Read-only
Dodge and burn slider (Split Grade)	Adds/removes exposure, ±2.25 stops	Drag right to darken, left to lighten

5.3 Step by step

1. Select the paper and filter brand.
2. Measure the dim highlight and the bright shadow, as in chapter 3.
3. Read the **SOFT EXPOSURE** and **HARD EXPOSURE** names and times.
4. Expose the print: soft leg first, then hard leg (or the reverse, if your workflow prefers; the total light is the same).

5. If the print needs adjustment, use the trims, then **Reset** to start again.
6. Use **DODGE & BURN** only for a final whole-print shift; it scales both legs.

> **Warning**
> The meter may warn that **highlights may burn** or that contrast is **limited**. Those warnings mean the requested split is outside what the paper can hold. Choose a less extreme **Midtones** position, or change the negative.

5.4 How the split is chosen, without the maths

The meter decides what share of the light each leg should carry. At one extreme the print is almost all soft; at the other, almost all hard. The **Midtones** slider moves that balance. The meter keeps the midtone anchored, so changing the balance does not make the whole print lighter or darker; it changes how the shadows and highlights separate.

The **Highlights** and **Shadows** trims adjust the two ends independently. The **Exposure** and **Contrast** trims adjust the whole print.

5.5 A worked example

A portrait negative:

- highlight **8.00 lux**;
- shadow **0.12 lux**;
- the range is about **6.1 stops**;
- the equivalent grade is **00**;
- soft leg: grade **00**, **29.5 s**;
- hard leg: grade **5**, **6.9 s**;
- total **36.4 s**.

The soft leg carries most of the light (about 89%) because the negative is very contrasty. If the highlights are still too dense, add a **Highlights** trim; if the shadows lose detail, add a **Shadows** trim.

5.6 Common questions

Why are there two times? The paper receives both exposures. The total is their sum; the split decides where the contrast goes.

Which leg first? It does not matter for the total exposure. Many printers do soft first, then hard; choose one order and keep it for consistency.

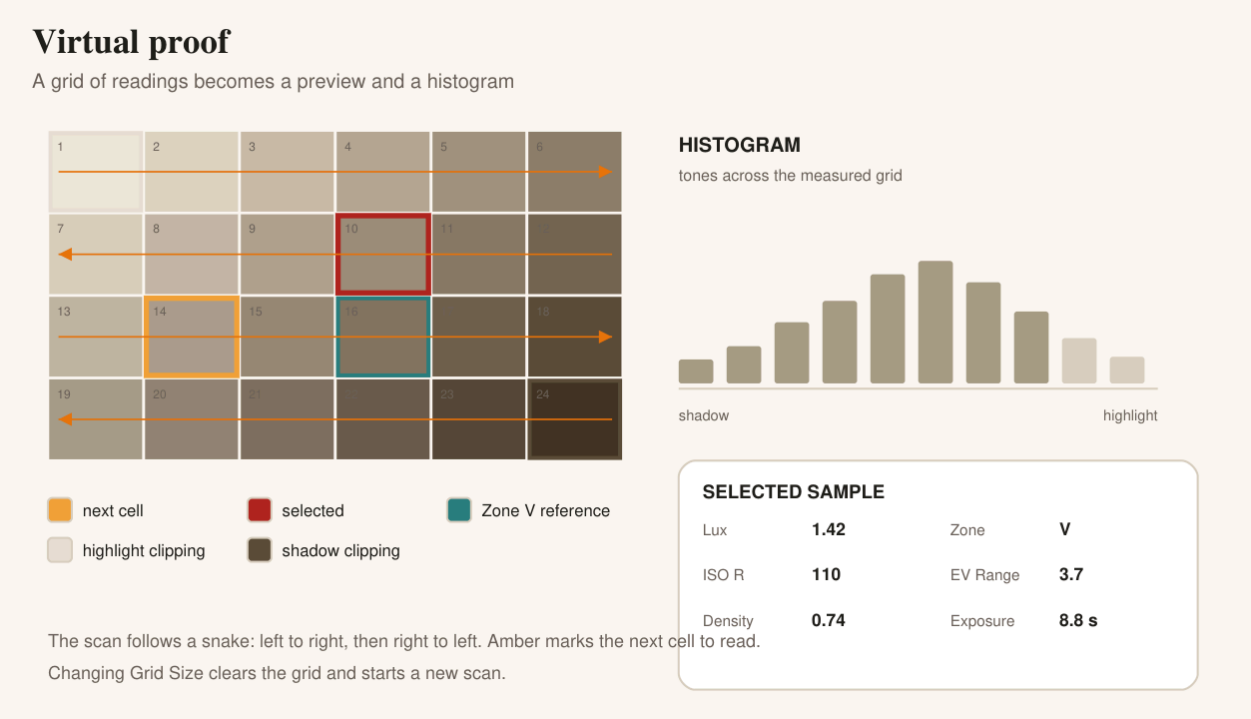
The soft and hard filters look odd. The names follow the selected **Filter Brand**; for FOMA papers a “Special” grade may appear as one of the legs. That is normal.

Can I use Split Grade on a fixed-grade paper? No. The tab will tell you it is not applicable. Use **Exposure**.

---

6. Virtual Proof

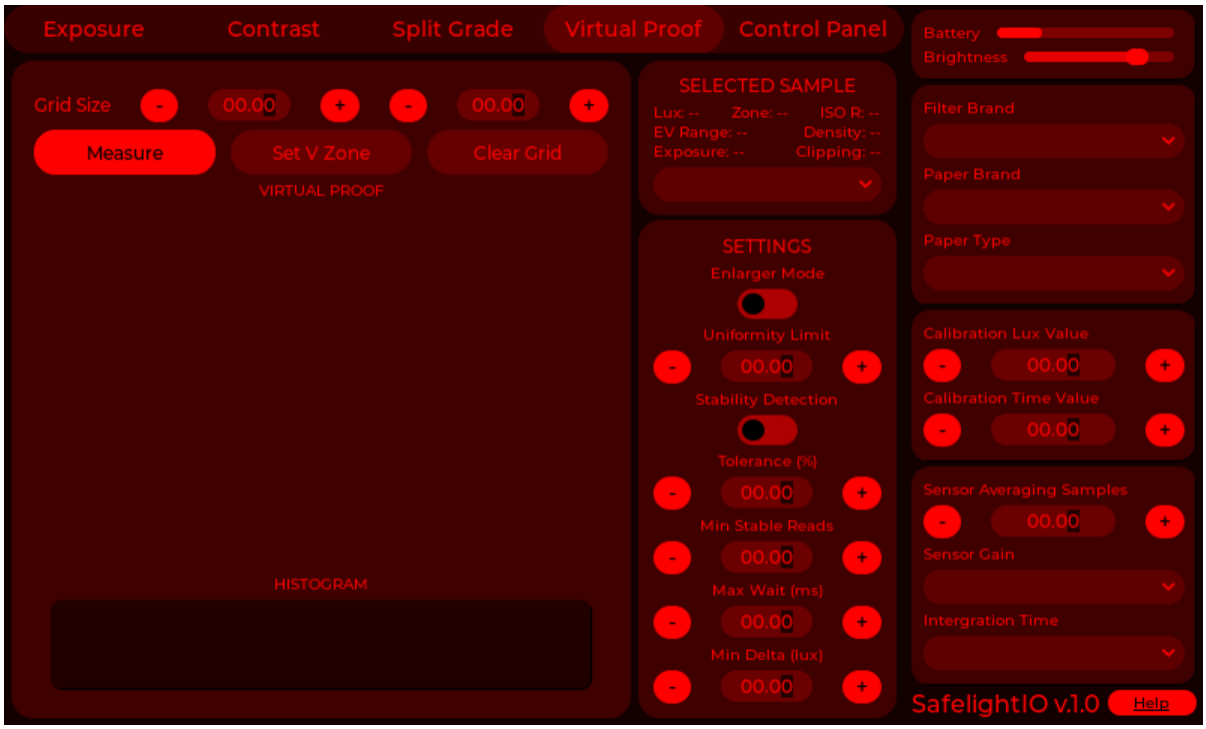
Use this tab when you want to see how the print will look across the easel before you commit a sheet.



6.1 What it does

The grid divides the easel into cells. You measure each cell; the device paints each cell with the gray it expects on the print at the selected paper and grade. The result is a low-resolution preview of the print, plus a histogram of the tones.

It is a proof, not a scanner: the preview is only as good as the cells you measured.



6.2 Every control and readout

Element	What it is	What to do
Grid Size	Sets the grid's width and height	Choose 3–5 cells per side for a quick proof; more cells for a detailed one
Grid width	Cells across, 2–40	Changing the size clears the grid
Grid height	Cells down, 2–30	Changing the size clears the grid
Measure	Reads the next cell	Press once per cell; the button becomes **Measure next (n/total)** during a scan
Set V Zone	Makes the selected cell the Zone V reference	Use it when you know which cell is the middle gray
Clear Grid	Erases all cells, the selection and the reference	Start a new proof
Grid canvas	The painted preview	Tap a cell to select it
HISTOGRAM	Distribution of the predicted gray values	Read the shape: left is shadow, right is highlight
SELECTED SAMPLE	The readouts for the selected cell	Tap a cell to inspect it
Lux (selected sample)	The measured light in that cell	Read-only
Zone (selected sample)	The predicted zone of that cell	Read-only
ISO R (selected sample)	The contrast range of the paper at this grade	Read-only
EV Range (selected sample)	The exposure range represented by the cell	Read-only
Density (selected sample)	The predicted print density	Read-only
Exposure (selected sample)	The exposure time for that cell	Read-only
Clipping (selected sample)	Whether the cell falls off the paper's range	If it clips, choose a different grade or split
SETTINGS	The proof's options	—
Filter grade selector (Virtual Proof)	The grade used to render the preview	Change it to see a different grade; the grid re-renders
Enlarger Mode switch	Turns the proof into a light-evenness test (section 6.5)	Use it with no negative in the carrier
Uniformity Limit	The allowed brightness variation in Enlarger mode, 5–50% (default 15%)	Set it to what your printing requires
Stability Detection switch	Turns on automatic settling during a scan	Use it for hands-free scans
Tolerance (%)	How close two reads must be to count as stable	Smaller is stricter
Min Stable Reads	How many stable reads are required	More is stricter
Max Wait (ms)	The longest the meter waits for a cell to settle	Prevents a scan from stalling
Min Delta (lux)	A floor for very dim cells	Stops noise from looking like change

> **Note**
> The four stability fields are used when **Stability Detection** is on. With it off, each press of **Measure** records a single reading immediately.

> **Note**
> When an automatic scan finishes, the meter picks a provisional Zone V reference: the cell nearest the middle of the measured brightness range in exposure terms. It is a starting point, not a guarantee that the cell is your middle gray. Select the cell you trust and press **Set V Zone** to take over. A manual, one-cell-at-a-time proof does not set a reference automatically.

6.3 The grid frames

The frame colour tells you what a cell is:

Frame	Meaning
Amber	The next cell the meter will read
Red	The cell you selected
Cyan	The Zone V reference cell
Light gray	Highlight clipping — the cell is too bright for the paper
Dark gray	Shadow clipping — the cell is too dark for the paper
No frame	An ordinary measured cell

Colour is a shortcut, not the only cue: the selected sample readouts and the histogram tell you the same story in numbers.

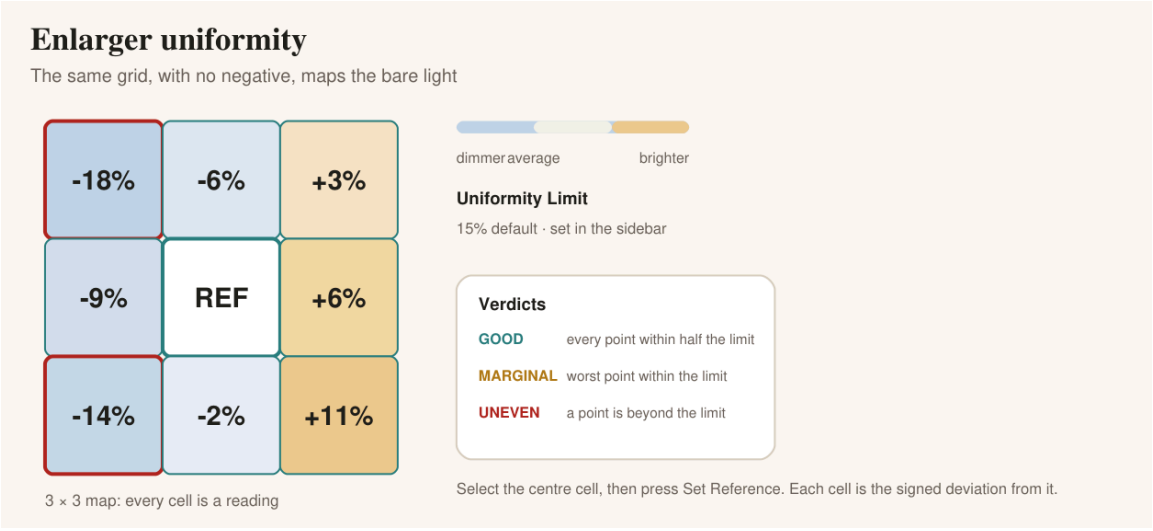
6.4 A proof in ten steps

1. Set **Grid Size** to a small grid, for example 4 × 3.
2. Choose the **Filter grade** you intend to print.
3. Press **Measure** for the first cell and place the sensor in the highlighted cell.
4. Keep pressing **Measure** (or turn on **Stability Detection** and let the meter scan) until the grid is filled.
5. Read the **HISTOGRAM**. A shape crowded on the right means the print will be bright; crowded on the left means dark.
6. Tap a cell to see its **SELECTED SAMPLE** readouts.
7. If you know the middle gray, select that cell and press **Set V Zone**.
8. Change the grade and watch the grid re-render.
9. **Clear Grid** to start again; changing **Grid Size** also clears it.
10. Save the session if the proof is worth keeping (chapter 9).

> **Tip**
> Six cells tell you whether the print is basically even. You do not need a 40 × 30 grid to catch a hot corner or a vignette.

6.5 Enlarger mode — is the light even?

Enlarger mode uses the same grid, but there is **no negative in the carrier**. The grid becomes a map of the bare enlarger light.



Set up:

1. Remove the negative. Leave the carrier empty.
2. Set the lens to the aperture and the head to the height you will print at.
3. Let the lamp warm up.
4. Put an opal diffuser over the sensor if you have one; do not let the meter shadow the point you are measuring.
5. Turn **Enlarger Mode** on. The filter grade selector is not used in this mode.
6. Set **Grid Size** to 3 × 3 (nine points) or 5 × 5 for a finer map.
7. Press **Measure** for each point, or turn on **Stability Detection** and scan.
8. Select the centre cell and press **Set Reference**. The centre is the reference for the deviation readouts.

The readouts:

Readout	Meaning
Lux	The light at the selected point
Dev %	How far the point deviates from the reference, in percent
Dev EV	The same deviation in stops
Range	The spread between the brightest and dimmest points
Mean	The average of all measured points
Min–Max	The dimmest and brightest points
Verdict	**GOOD**, **MARGINAL** or **UNEVEN**

The verdict:

- **GOOD** — every point is within half the **Uniformity Limit**.
- **MARGINAL** — the worst point is between half the limit and the full limit.
- **UNEVEN** — the worst point is beyond the limit.

The default limit is **15%**, which is roughly 0.2 stop. That is a reasonable working tolerance for most enlargers and papers. A stricter limit (for example 10%) is useful for critical work; a looser one (25–30%) may be acceptable for rough proofs.

What the map tells you:

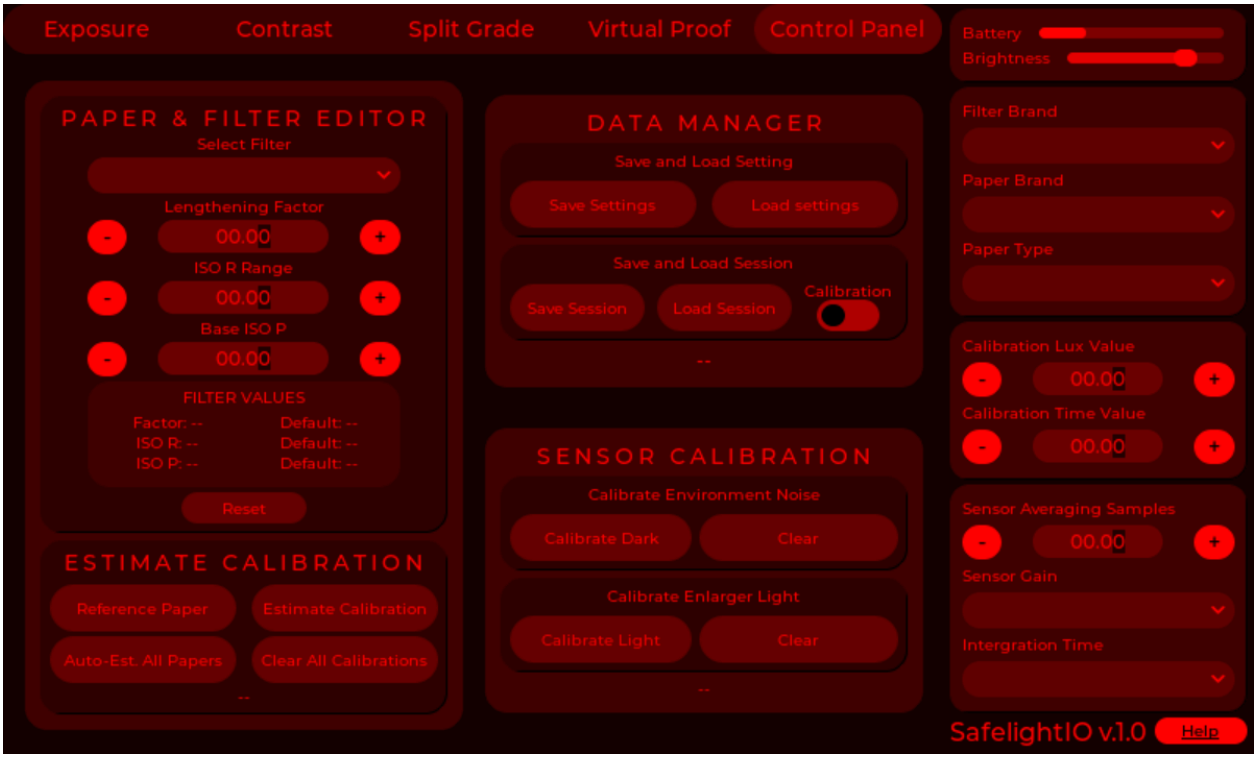
- dark corners — condenser or diffuser misalignment, or a lens that does not cover the format;
- a bright centre with dark edges — normal falloff for a condenser enlarger, worse at large magnification;
- an off-centre hot spot — the lamp is not centred;
- a single dark cell — something is shadowing that point (your hand, the meter, a cable).

Enlarger mode measurements are kept when you switch back to negative mode, so you can inspect them later. Saving the session stores the mode and the grid (chapter 9).

7. Control Panel

Use this tab to edit filter data, estimate calibrations, save/load, and calibrate the sensor.

The tab is labelled **Control Panel**. Its main heading is **PAPER & FILTER EDITOR**. The Help Center calls the same area **Paper Editor**. The tab has four sections in two columns:



- **PAPER & FILTER EDITOR** — left column, top. Override the built-in factors and ISO values.
- **ESTIMATE CALIBRATION** — left column, bottom. Copy a calibration to other papers by speed.
- **DATA MANAGER** — right column, top. Save and load settings and sessions.
- **SENSOR CALIBRATION** — right column, bottom. Dark and light calibration.

7.1 Paper & Filter Editor

This section changes the numbers the meter uses for a **paper + filter grade** combination. It is for advanced users; the built-in data is good enough for most printing. Edits are saved in flash and survive a power cycle.

Element	What it is	What to do
Select Filter	Chooses the grade to edit	Pick the grade whose data you want to change
Lengthening Factor	The filter's exposure multiplier	Raise it if prints through that filter are too light; lower it if too dark
ISO R Range	The contrast range this grade produces	Higher = softer contrast; lower = harder
Base ISO P	The paper's unfiltered speed	Higher = shorter times; lower = longer times. This is per paper, not per grade
FILTER VALUES	The read-only value block	—
Factor value	The **effective** factor used by every tab	Shows your override, or the built-in value
Default factor value	The built-in factory factor	Reference only
ISO R value	The **effective** ISO R	Shows your override, or the built-in value

Default ISO R value	The built-in ISO R	Reference only
ISO P value	The **effective** ISO P for the paper	Shows your override, or the built-in value
Default ISO P value	The built-in ISO P	Reference only
Reset	Clears all overrides for the **current paper**	Other papers keep their overrides

- A factor of **1.00** means "no exposure change".
- A readout shows `--` when there is no value to display yet, for example before a grade is selected.
- Two decimal places are enough for the factor; ISO R and ISO P are whole numbers.
- The device stores at most **64 overrides**. If the list is full, reset a paper you no longer use.
- On a fixed-grade paper, Factor and ISO R are disabled; only ISO P can be changed.
- Changing a value commits that field. Selecting another grade commits a pending edit first.

> **Note**
> **ISO R** is the paper's contrast range: a higher number is a softer paper. **ISO P** is its speed: a higher number means a faster paper and a shorter time. You do not need the standards to use the meter; you only need to know which way the numbers move.

7.2 Estimate Calibration

This section copies a calibration from one paper to another. It is a shortcut for papers you have not calibrated yet; the result is an estimate, not a measurement.

Element	What it is	What to do
ESTIMATE CALIBRATION	The section heading	—
Reference Paper	Snapshots the current paper's calibration (lux, time, ISO P)	Calibrate one paper well, select it, then press **Reference Paper**
Reference status	Tells you whether a reference is stored	If it says **No reference**, set one first
Estimate Calibration	Estimates the current paper from the reference, scaled by ISO P	Use it for a paper you have not calibrated
Auto-Est. All Papers	Estimates every uncalibrated paper from the reference	Asks for confirmation first
Clear All Calibrations	Wipes every paper calibration and the reference	Asks for confirmation first; this cannot be undone

The estimate assumes that, at the same light level, a paper with twice the ISO P needs half the time. It is a starting point: make a test strip and adjust.

A sensible workflow:

1. Calibrate your most-used paper properly (chapter 10).
2. Select it and press **Reference Paper**.
3. Select an uncalibrated paper and press **Estimate Calibration**.
4. Print a test strip; if the time is off, calibrate that paper properly or adjust its factor.
5. Use **Auto-Est. All Papers** only when you want rough starting points for the whole collection.

7.3 Data Manager

Element	What it is	What to do
DATA MANAGER	The section heading	—

Save and Load Setting	The settings group	—
Save Settings	Writes the internal flash snapshot and `BACKUP.TXT`	Do it before a firmware update
Load settings	Restores the flash snapshot; reboots	Asks for confirmation first
Save and Load Session	The session group	—
Save Session	Writes the current readings and results to `SESSION.CSV`	Use it to resume later or share a setup
Load Session	Restores a session	Asks for confirmation if measurements exist
Calibration switch (session load)	Include or exclude the stored calibration when loading	On if you want the session's calibration restored; off to keep the current one
Data manager message	The result line: saved, loaded, **Nothing to export**, **No session file**	Read it if a button seems to do nothing

A **session** contains the current paper and filter, the last Contrast and Split readings and results, and the Virtual Proof grid (including its mode). It does not contain your whole settings file; use **Save Settings** for that.

7.4 Sensor Calibration

Element	What it is	What to do
SENSOR CALIBRATION	The section heading	—
Calibrate Environment Noise	The dark-calibration group	Do it in total darkness
Calibrate Dark	Measures the sensor's electrical noise floor	Press once in darkness; it sweeps several settings automatically
Clear (dark)	Erases the dark calibration	Asks for confirmation
Sensor calibration status	Progress and result, for example `dark calib... 12/24` or `dark offsets set (24)`	Wait for it to finish
Calibrate Enlarger Light	The light-calibration group	Do it with the enlarger on and steady
Calibrate Light	Measures the gain stages against a known light level	Follow the two-phase prompt
Clear (light)	Erases the light calibration	Asks for confirmation

Calibrate Dark

1. Put the sensor in **total darkness** — cap it, or close the enlarger and turn off all light.
2. Press **Calibrate Dark**.
3. The device sweeps 24 combinations of gain and integration time. It shows `dark calib... n/24` while it works; at the slowest settings this takes a few minutes.
4. Wait for `dark offsets set (24)`. If it says `dark calib failed`, light reached the sensor; repeat in darkness.

Calibrate Light

1. Put the sensor in **steady enlarger light** — no negative, lens open, lamp warmed up. Use a mid-range light level, not the brightest the enlarger can produce.
2. Press **Calibrate Light**. The device starts phase 1 and shows `gain calib... phase 1/2`.
3. When it asks, dim the light by about **four stops** (close the aperture or use a neutral density filter) and choose **Yes**.
4. Wait for `gain cal set`. If it says `gain cal failed - adjust light`, the level was too high or too low; adjust and repeat.
5. The result is stored. If the light moved during the calibration, it reports `- moved`; if there was not enough infrared, `- no IR`; if a stage saturated, `- MAX sat` or `HIGH zero`.

Run the light calibration once per device. Repeat it if readings disagree when you switch gain, or after a firmware update that changes the sensor settings.

> **Important**
> A safelight, a phone screen or a gap in the enlarger can spoil the dark calibration. A flickering lamp or a moving hand can spoil the light calibration. Both are worth doing carefully once.

---

8. Settings sidebar

The sidebar is always visible. It holds the choices that affect every tab. From top to bottom the panels hold:

- battery and **Brightness**;
- **Filter Brand**, **Paper Brand** and **Paper Type**;
- **Calibration Lux Value** and **Calibration Time Value**;
- **Sensor Averaging Samples**, **Sensor Gain** and **Intergration Time**;
- the battery status line, the version and **Help**.



Element	What it is	Range / default	What to do
Battery + bar	Charge state	0–100%	See 8.4
Brightness + slider	Screen backlight	1–100	Set the dimmest level you can read; saves battery and stray light
Filter Brand	The filter system in your enlarger	9 brands	Choose the brand printed on your filters
Paper Brand	The paper manufacturer	Built-in list	Choose before measuring
Paper Type	The paper surface/emulsion	Built-in list	Choose before measuring

Calibration Lux Value	The lux reading from your calibration test strip	0.00–99.99	Enter it during paper calibration
Calibration Time Value	The exposure time that produced the good test strip	0.00–99.99	Enter it during paper calibration
Sensor Averaging Samples	How many readings are averaged per measurement	1–16, default 5	Raise for dim or flickering light; lower for speed
Sensor Gain	Sensitivity stage	Auto, LOW, MED, HIGH, MAX; default Auto	Leave on Auto unless you need a fixed stage
Intergration Time	How long each sensor reading lasts	100–600 ms, default 200	Longer is quieter and slower; shorter is faster
SafelightIO v.1.0	Firmware version	—	Use it when checking an update
Help	Opens the Help Center	—	Quick reference on the device

8.1 Choosing paper and filter

Paper Brand and **Paper Type** are linked: choose the brand first, then the paper. Selecting a paper loads its calibration, its trims and its filter data. If you change paper, measure again; the previous readings belonged to the previous paper.

Filter Brand changes the grade names and the built-in factors. It does not change the paper. Choose the brand of the filters you actually own.

8.2 Sensor gain, integration time and samples

These three controls trade speed against noise:

- **Sensor Gain**: Auto starts at the most sensitive stage and steps down when the light is bright enough. A fixed stage is useful when you want repeatable readings and you know the light level; if a fixed stage saturates, the reading shows `error`.
- **Intergration Time**: the length of one reading. Longer readings collect more light and are less noisy; shorter readings make a proof scan faster.
- **Sensor Averaging Samples**: how many readings are averaged. More samples make a steadier number; fewer make the device feel quicker.

For a Virtual Proof scan, lower the integration time and sample count. For a single critical reading, raise them.

8.3 The calibration values

Calibration Lux Value and **Calibration Time Value** are the two numbers from your paper calibration test strip. The device divides one by the other to get the paper's constant. You only touch them during calibration (chapter 10).

8.4 Battery, charging and power

The battery bar is red and updates every five seconds. The device has no charger status pin, so it infers charging from the battery voltage:

Display	Meaning
Discharging + percentage	Running on the battery
Charging... (bar hidden)	USB-C connected; the device is charging
Fully charged	The cell has reached the charger's top voltage
Battery absent (grey, empty)	No battery detected; USB-C power only
Battery low	About 10% or less; charge now

Notes:

- **Battery low** replaces the bar when the charge reaches about 10%. The device has no sleep mode, so the warning stays visible until you charge it.
- **Charging…** can stay on after the battery is effectively full, because some chargers stop just below the top voltage the device looks for. This is normal.
- The bar is voltage-based. Under a heavy load it can sag; a bar that looks empty while the device still runs is not necessarily a dead battery.
- Charge with USB-C. The battery is protected; unplug when you are done printing.
- **Brightness** at 1 is intentionally very dim, not off. Drag right if the screen looks dead.

---

9. Saving, loading and storage

The meter can keep its data in two places:

- **Internal flash** — always present. Settings, calibrations and trims live here.
- **microSD card** — optional. It holds copies and files you can move between devices or edit on a computer.

File	Where	What it is
`SETTINGS.TXT`	SD card	The settings file
`SETTINGS.BAK`	SD card	The previous settings file
`PAPERS.CSV`	SD card	Custom paper definitions
`BACKUP.TXT`	SD card	A readable backup of the settings
`SESSION.CSV`	SD card	One saved session

9.1 Settings and backups

- **Save Settings** writes the internal flash snapshot and `BACKUP.TXT`. Do this before a firmware update.
- **Load settings** restores the snapshot and reboots. It asks for confirmation.
- Settings changes are flushed about half a second after the last edit. If you cut power immediately after a change, the last change may be lost.
- With a card inserted, the device also keeps `SETTINGS.TXT` on the card. On first boot with a card it copies the current settings to the card. If the card is removed, the device falls back to flash.
- Removing the card is not detected instantly; the sidebar can still say `SD: settings.txt` until the next save fails. If a save fails, reinsert the card and try again.

9.2 Sessions

A session is a snapshot of **the current work**, not the whole meter:

- the selected paper and filter;
- the last Contrast and Split readings and results;
- the Virtual Proof grid, its reference and its mode (negative or enlarger);
- optionally the paper calibration.

Save Session writes `SESSION.CSV`. **Load Session** restores it, re-derives the results and asks before overwriting current measurements. The **Calibration** switch decides whether the session's calibration is restored too.

Sessions are useful when you stop halfway through a print, when you want to compare two negatives, or when you want to send a setup to another meter.

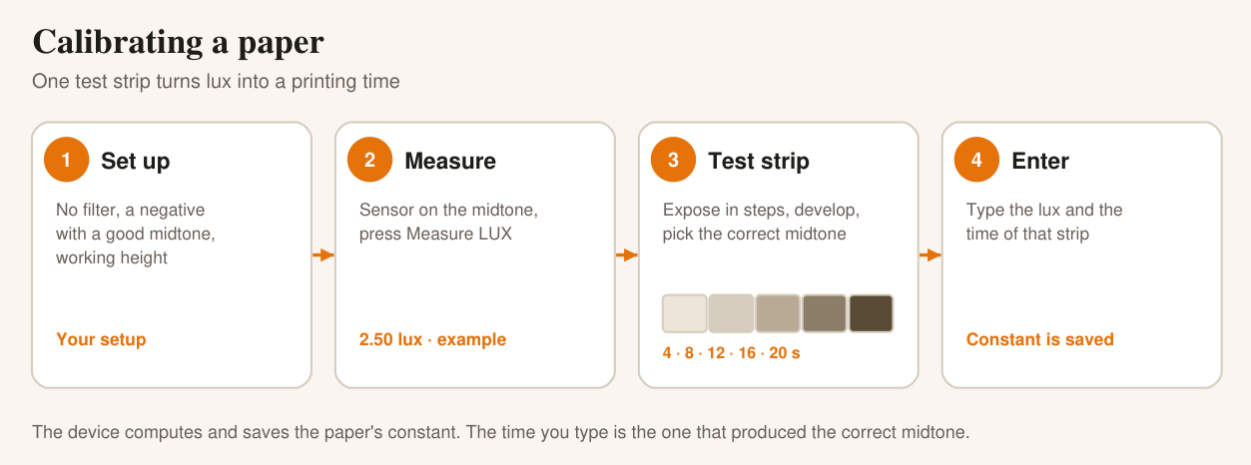
9.3 What is kept, and what is not

- Settings, paper calibrations, filter overrides and trims live in internal flash.
- Custom paper **definitions** live in `PAPERS.CSV` on the card. Their calibrations are stored on the card too; without the card they are not available after a reboot.
- `BACKUP.TXT` is a readable copy, not a second live database.
- A firmware update preserves the settings record only while the settings format is unchanged. A format change resets the stored settings; the release notes will say so. Recalibrate the sensor and your papers after such an update.

---

10. Calibrating a paper

Do this once per paper. It is the step that turns lux into seconds.



10.1 Before you start

- The enlarger is at the height, aperture and lens you will use for real prints.
- The paper is the one you will print on.
- The sensor is calibrated (section 7.4).
- You have a test strip and your usual developer.

10.2 Procedure

1. **Choose the paper** in the sidebar.
2. **Set up a normal print.** Use a negative with a good midtone and no contrast filter. Focus at the working height.
3. **Place the sensor** on the easel where the midtone will fall. Do not let the meter or your hand shadow the point.
4. **Press Measure LUX** and write down the lux reading.
5. **Make a test strip** at a few seconds, then in steps, through the same lens and aperture. Develop it normally.
6. **Choose the strip with the correct midtone.** Note its exposure time.
7. In the sidebar, enter the lux reading in **Calibration Lux Value** and the time in **Calibration Time Value**.
8. The device computes the constant. Open **Exposure** and check that it now shows a time instead of `calibrate`.
9. Make a full print to confirm. If it is consistently off by a small amount, use the **Exposure** trim rather than recalibrating.

10.3 What “correct midtone” means

The test strip should show the tone you want for a middle gray — not the lightest strip that still has detail, and not the darkest. The meter uses that point as Zone V. If you choose a different tone, the constant will be biased.

10.4 Redoing a calibration

Repeat the procedure and enter the new lux and time. The new values overwrite the old ones for that paper. You do not need to clear anything first.

Calibrate again if:

- you change the enlarger, lens or aperture you print at;
- you change the paper's developer or dilution significantly;
- the prints are consistently off by a fixed factor and a trim no longer covers it;
- you replace the sensor.

10.5 Fixed-grade papers

Fixed-grade papers have one contrast grade. The calibration procedure is the same; just use the paper's grade (no variable-contrast filter). The Contrast tab will show **Fixed grade** and Split Grade is not applicable.

10.6 A calibration checklist

- [ ] Sensor dark calibration done.
- [ ] Sensor light calibration done.
- [ ] Paper selected in the sidebar.
- [ ] Same lens, aperture and height as real prints.
- [ ] Sensor on the midtone, not shadowed.
- [ ] Lux reading written down.
- [ ] Test strip developed normally.
- [ ] Correct midtone chosen.
- [ ] Lux and time entered.
- [ ] Exposure tab shows a time.

10.7 Worked examples

These examples follow the original guide's test-strip method. The first uses its numbers; the others show common variations.

Example 1 — the first calibration

You are printing on **Ilford Multigrade RC Deluxe** at f/8, with the head at your usual height and no contrast filter.

1. You measure **2.50 lux** on the midtone.
2. You make a test strip in 4-second steps: **4, 8, 12, 16, 20 s**.
3. The **12-second** strip has the middle gray you want.
4. Enter **2.50** in **Calibration Lux Value** and **12.0** in **Calibration Time Value**. The device stores **30.0** lux-seconds (2.50 × 12.0).
5. With the same negative and no changes, press **Measure LUX** again. It should read about **2.50 lux**, and the Exposure tab should show **12.0 s**. That is the check that the calibration is good.

Example 2 — recalibrating after a developer change

You mix a new developer and the same paper now prints darker at the same time.

1. The light reading is still **2.50 lux**.
2. The test strip says the correct midtone is now **9.0 s**, not 12.0 s.
3. Enter **2.50** and **9.0**. The constant becomes **22.5 lux-seconds**.
4. The paper is now calibrated for this developer. The old value is overwritten; there is nothing to clear first.

The constant changed because the paper/developer combination changed. That is expected. If only the enlarger height or aperture changed, the light reading changes instead, and the constant should stay close to the old value.

Example 3 — a fixed-grade paper

You are printing on **FOMA FOMASPEED Normal**, which has one contrast grade.

1. Select the paper. Use no contrast filter.
2. Measure **2.50 lux** on the midtone.
3. The test strip gives **12.0 s**. Enter **2.50** and **12.0**.
4. The Exposure tab gives a time. The Contrast tab shows **Fixed grade**, and Split Grade is not applicable.

The procedure is the same as for a variable-contrast paper. The only difference is that there is no grade to choose.

Example 4 — a second paper from a reference

You have already calibrated **Ilford MG FB Classic** at **1.42 lux** and **8.8 s**. You want a starting point for **FOMA FOMABROM Variant**.

1. On the calibrated paper, press **Reference Paper**.
2. Switch to **FOMA FOMABROM Variant**.
3. Press **Estimate Calibration**.
4. The meter scales the reference by the paper speed (ISO P 500 → 130), about **×3.85**, and suggests **33.9 seconds**.

This is a starting point, not a measurement. Make a real test strip before you trust it for critical work.

Example 5 — no strip looks right

The 12-second strip is too dark and the 8-second strip is too light.

- Make a new strip through the same setup at **9.5, 10.0, 10.5 and 11.0 s**.
- If even the whole strip is too dark or too light, move the range by a full stop and repeat. Do not guess the constant.
- When one strip is clearly right, enter its time. The constant is what you measured, not what you hoped for.

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11. The paper and filter database

11.1 Papers

The built-in database has **22 papers** across Ilford and FOMA. Most are variable-contrast; **eight are fixed-grade**. The Paper Type dropdown is the source of truth for the exact list on your firmware.

Fixed-grade papers:

- FOMA FOMASPEED Normal
- FOMA FOMASPEED Hard
- FOMA FOMABROM Normal
- FOMA FOMABROM Hard
- FOMA FOMABROM N chamois, glossy
- FOMA FOMABROM N chamois, matt
- FOMA RETROBROM Special, glossy
- FOMA RETROBROM Special, semi-matt

The variable-contrast papers include the Ilford Multigrade RC and FB families (Deluxe, Portfolio, Warmtone, Cooltone, Classic, ART 300) and the FOMA Variant, Fomabrom Variant, Fomapastel and Fomatone families. Some entries are marked **~** in the database documentation because their values are derived from similar papers rather than measured. They are reasonable starting points; calibrate the paper for critical work.

11.2 Filter brands

The **Filter Brand** list has **nine** entries:

Brand	Grade names you may see
Ilford	00, 0, 1, 2, 3, 4, 5
FOMA	Y, M, and Special variants
AGFA	00–5
KODAK	00–5
DURST	00–5
MEOPTA	00–5
BESELER	00–5
OMEGA/CHROMEGA	00–5
DE VERE	00–5

The exact grades offered depend on the paper and the brand. If a grade is not available for the selected paper, it does not appear in the list.

11.3 Adding your own papers

Custom papers are read from `PAPERS.CSV` on the microSD card at boot. You do not need to rebuild the firmware.

The file has a header line and two kinds of rows:

- a **P** row defines a paper (id, manufacturer, type, contrast type, family, Dmin, Dmax, fibre base, ISO P);
- one or more **F** rows define that paper's filter grades (grade, factor, ISO R).

Rules:

- Commas or semicolons separate fields.
- A **P** row needs at least 10 fields; an **F** row needs at least 8.

- Up to 8 filter rows per paper.
- A paper whose id matches a built-in replaces it; other ids are added at the end.
- The file is a template on first boot with a card. Edit a copy, keep the header, and reboot with the card inserted.
- Calibrations for custom papers live on the card; without the card they are not available after a reboot.

The exact column order is documented in the header comment that the device writes into the template. Keep a backup of the file; it is not part of `BACKUP.TXT`.

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12. Filter factors

12.1 What a filter factor is

A contrast filter absorbs light. The **factor** is how many times longer the exposure must be to compensate. A factor of 2.0 doubles the time; 1.50 adds half.

The meter has built-in factors for each paper and grade. They are manufacturer averages; your enlarger, lamp and filters can differ by 10–30%. If your prints through a filter are consistently light or dark, adjust the factor in the Control Panel (chapter 7.1).

12.2 Measuring your own factor, simply

1. Without a filter, make a test strip of a midtone and find the correct time.
2. With the filter you want to measure, make another strip of the same subject at the same aperture and height.
3. Divide the filtered time by the unfiltered time. That is the factor.
4. Enter it in **Lengthening Factor** for that grade.

Example: 12.0 s unfiltered and 26.4 s filtered → 26.4 ÷ 12.0 = **2.20**.

You do not need a densitometer. A consistent test strip is enough.

12.3 Re-measuring

Re-measure after changing the lamp, reflector, lens or filter set; when prints through one grade are consistently off; or once a year if you print often.

12.4 Long exposures

At very long exposures, paper loses some sensitivity (reciprocity failure). The meter applies a correction automatically when the time is long enough, and the **Exposure Note** tells you when it has done so. You do not need to calculate it. If you override a filter factor, use the measured value; an incorrect factor compounds with the long-exposure correction.

13. Messages, Help Center and troubleshooting

13.1 What the messages mean

The device uses short messages because the screen is small. Here is the full list.

Message	Where	Meaning	What to do
`calibrate`	Exposure time	The selected paper has no calibration	Calibrate the paper (chapter 10)
`measure lux`	Exposure time	The paper is calibrated, but there is no live reading	Press **Measure LUX**
`no sensor`	Lux readout	The sensor is not responding	Check the sensor cable; reboot
`error`	Lux readout	A reading failed, usually saturation at a fixed gain	Choose **Auto** gain, or a lower fixed gain
`read error`	Lux readout	The sensor reading failed	Reboot; if it persists, check the cable
`Fixed grade`	Contrast tab	The selected paper has one contrast grade	Use Exposure; Split Grade is not applicable
`--`	Contrast / Split	One of the two readings is missing, or the paper is not calibrated	Read the note; measure the missing point
`dark calib... n/24`	Sensor status	Dark calibration in progress	Wait
`dark offsets set (24)`	Sensor status	Dark calibration complete	—
`dark calib failed`	Sensor status	Light reached the sensor during calibration	Repeat in total darkness
`gain calib... phase 1/2`	Sensor status	Light calibration, first phase	Wait for the prompt
`gain cal set`	Sensor status	Light calibration complete	—
`gain cal failed - adjust light`	Sensor status	The light level was outside the usable range	Adjust the enlarger and repeat
`- moved`	Sensor status	The light changed during calibration	Keep the light steady and repeat
`- no IR`	Sensor status	Not enough infrared for the calibration	Use a tungsten/halogen enlarger light, or accept the limitation
`- MAX sat` / `HIGH zero`	Sensor status	One gain stage could not be calibrated	The other stages are still usable; repeat if needed
`No reference`	Estimate calibration	No reference paper has been set	Calibrate a paper, select it, press **Reference Paper**
`Nothing to export`	Data manager	There is no session data to save	Take a reading first
`No session file`	Data manager	No `SESSION.CSV` on the card	Save a session first
`SD: none (flash)`	Sidebar / data	No card; settings are in internal flash	Insert a card if you want file storage
`Battery low`	Battery area	About 10% charge or less	Charge with USB-C
`Charging...`	Battery area	USB-C connected, charging	—
`Fully charged`	Battery area	The cell reached the top voltage	Unplug when convenient
`Battery absent`	Battery area	No battery detected	Use USB-C power, or fit the battery

> **Note**
> The on-screen strings use `...` and `-`, not typographic ellipses or dashes. This manual quotes the exact string so you can match it.

13.2 The Help Center

Press **Help** at the bottom of the sidebar. The Help Center opens with a **Back** button, the version, and seven tabs:

Help tab	Covers
Start	First-time setup, the one idea, what you need, a QR code to the project page
Settings	The sidebar: paper, filter, sensor, brightness, battery
Exposure	The Exposure tab
Contrast	The Contrast tab
Split Grade	The Split Grade tab
Virtual Proof	The proof grid, stability and Enlarger mode
Paper Editor	The Control Panel's paper/filter editor

Each tab has a short explanation and a picture of the screen. The Help Center is a reminder for when you are at the enlarger; this manual is the full reference. The pictures in the Help Center may show an earlier firmware build; the text in this manual matches firmware 1.0.

The **Start** tab also shows a QR code for the project page. The address is `https://atriantas.github.io/SafelightIO/`.

13.3 Troubleshooting

The meter gives no time

Symptom	Likely cause	Fix
Exposure time shows `calibrate`	Paper not calibrated	Chapter 10
Exposure time shows `measure lux`	No reading yet	Press **Measure LUX**
Lux shows `no sensor`	Sensor not responding	Check the cable; reboot
Lux shows `error`	Saturation at a fixed gain	Set **Sensor Gain** to **Auto**, or lower the fixed gain

The time is consistently wrong

Symptom	Likely cause	Fix
Every print is 2× too light or too dark	Enlarger height or aperture differs from calibration	Recalibrate the paper at the height/aperture you print at
Every print is 2× too light or too dark, at the same setup	Gain stages uncalibrated	Run **Calibrate Light**
Readings disagree when you change gain	Gain stages uncalibrated	Run **Calibrate Light**
One paper is off, others are fine	That paper's calibration	Recalibrate that paper
Long exposures are off	Reciprocity correction or a stale filter factor	Check the **Exposure Note**; remeasure the filter factor

Contrast and Split Grade

Symptom	Likely cause	Fix
`--` everywhere	A reading is missing or the paper is uncalibrated	Measure both points; calibrate the paper
Grade looks too hard/soft	Match Quality is **Approximate** or a trim is active	Check the trims; press **Reset**
Fixed grade shown	Fixed-grade paper selected	Use Exposure; Split Grade is not applicable
Highlights burn every time	Negative range exceeds the paper	Choose a softer grade, or use Split Grade

Contrast grade does not change	Filter Brand or paper limits the available grades	Check the selected brand; not every brand offers every grade

Virtual Proof

Symptom	Likely cause	Fix
Grid is empty after changing size	Changing Grid Size clears the grid	Rescan
Every highlight is clipped	The negative is too contrasty for the paper/grade	Choose a softer grade, or use Split Grade
A cell will not settle	Light is flickering or the sensor is moving	Keep the light steady; raise **Max Wait** or turn off **Stability Detection**
Set V Zone seems to change everything	It changes the reference for the whole grid	Use it deliberately; Clear Grid resets it
Enlarger verdict is UNEVEN	Light falls off or is misaligned	See section 6.5

Control Panel and data

Symptom	Likely cause	Fix
Filter editor shows stale values	The spinboxes follow the grade dropdown	Re-select the filter grade
Reset did nothing	Reset only affects the current paper	Select the paper you want to reset
Estimate Calibration says No reference	No reference paper	Set **Reference Paper** first
Clear All Calibrations wiped everything	You confirmed the action	Recalibrate; the confirmation screen is deliberate
Save Session says Nothing to export	No readings yet	Take a reading first
Load Session says No session file	No `SESSION.CSV` on the card	Save a session first
Settings changes lost after power-off	Flush takes about half a second	Wait a moment after the last edit
Custom papers gone after reboot	Card removed or file not read	Insert the card and reboot

Screen, power and sensor

Symptom	Likely cause	Fix
Screen looks almost black	Brightness at 1	Drag **Brightness** right
Screen dims during Measure	The backlight dims to 6% during reads	Normal; it restores afterwards
Screen dark for several seconds	Long sample count at 600 ms	Reduce samples or integration time
Battery bar empty but device runs	Voltage sag under load	Charge; **Battery low** is the real warning
Charging... never ends	Charger stops just below the top voltage	Normal
The device does not sleep	There is no sleep mode	Use **Brightness** and charge when needed

> **Tip**
> Check the sensor cable, the selected paper, and the battery. The three cause most “wrong reading” reports.

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14. Firmware updates

The firmware can be updated over USB-C with a `.uf2` file. No special tools or drivers are needed.

14.1 Check your version

The firmware version is **v1.0**. You can see it:

- on the splash screen (**SafelightIO Darkroom Meter v.1.0**);
- at the bottom of the sidebar (**SafelightIO v.1.0**);
- in the Help Center.

14.2 Update

1. **Get the file.** Use the firmware file supplied for your meter. The project page is `https://atriantas.github.io/SafelightIO/`; the QR code on the Help Center's Start tab opens it.
2. **Enter the bootloader.** With the meter off, press the small **BOOTSEL** hole at the top of the device with a pin, keep it pressed, and connect USB-C. The computer shows a removable drive called **RP2350**.
3. **Copy the `.uf2` file** to that drive. The meter verifies it and reboots.
4. **Wait for the main screen.** Your settings and paper calibrations are kept unless the release changes the settings format.

> **Warning**
> Do not install a firmware file for a different device. If the drive does not appear, try a data-capable USB-C cable and press the BOOTSEL hole again.

14.3 If the update does not start

Symptom	Likely cause	Fix
No **RP2350** drive	The BOOTSEL hole was not pressed, or the cable is charge-only	Press the hole with a pin and repeat with a data cable
The file disappears immediately	The file is corrupt or for another device	Download it again from the project page
The meter still runs the old version	The copy did not complete	Repeat the update
Calibrations are gone	The settings format changed	Recalibrate the sensor and your papers

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15. Reference

15.1 Ranges and defaults

Setting	Range	Default
Zone of measurement	0–X	V
Calibration Lux / Time	0.00–99.99	—
Sensor Averaging Samples	1–16	5
Sensor Gain	Auto, LOW, MED, HIGH, MAX	Auto
Intergration Time	100–600 ms	200 ms
Brightness	1–100	—
Grid width	2–40	—
Grid height	2–30	—
Uniformity Limit	5–50%	15%
Filter factor	0.01–99.99	built-in per grade
ISO R / ISO P	1–999	built-in per paper
Filter overrides	up to 64	0
Contrast/Split trims	±1.5 stops, 1/6-stop steps	0
Dodge & burn	±2.25 stops, ¼-stop steps	0
Battery warning	10%	—
Battery poll	every 5 s	—

15.2 Glossary

- **Zone** — a step on the print's tone scale, from 0 (black) to X (paper white). Zone V is middle gray.
- **Grade** — the contrast of the paper/filter combination. Higher grade = harder (more contrast).
- **Stop / EV** — a doubling or halving of exposure. One stop = 1 EV.
- **Contrast Range** — the brightness range of the negative, in stops.
- **Highlight** — the brightest important part of the print; it comes from the **dim** part of the projected image.
- **Shadow** — the darkest important part with detail; it comes from the **bright** part of the projected image.
- **Midtone** — the middle of the tone range; the meter anchors exposure there.
- **Calibration constant** — the paper's lux × seconds value. It turns a light reading into a time.
- **ISO R** — the paper's contrast range. Higher = softer.
- **ISO P** — the paper's speed. Higher = faster, shorter time.
- **Filter factor** — how many times longer the exposure is through a filter.
- **Dodge / burn** — hold back or add light locally under the enlarger. The device's **DODGE & BURN** panel is a whole-print trim.
- **Split grade** — two exposures through a soft and a hard filter.
- **Virtual proof** — a grid preview built from point readings.
- **Uniformity** — how evenly the enlarger lights the easel.
- **Reciprocity** — the loss of paper sensitivity at very long exposures; the meter corrects for it automatically.

- **Lux** — the unit of illuminance the sensor measures.
- **Fixed grade** — a paper with one contrast grade; no variable filter.
- **Session** — a saved snapshot of the current readings, grid and settings.
- **Reference paper** — the calibrated paper used as the source for estimates.

15.3 Index of controls

Control or readout	Where
ADDITIONAL TIME	Exposure, Contrast, Split Grade
ANALYSIS DETAILS	Split Grade
Auto-Est. All Papers	Control Panel → Estimate Calibration
Back	Help Center
Base ISO P	Control Panel → Paper & Filter Editor
Battery	Sidebar
Brightness	Sidebar
CALIBRATION CONSTANT	Exposure, Contrast, Split Grade
Calibration Lux Value	Sidebar
Calibration Time Value	Sidebar
Calibrate Dark	Control Panel → Sensor Calibration
Calibrate Enlarger Light	Control Panel → Sensor Calibration
Calibrate Environment Noise	Control Panel → Sensor Calibration
Calibrate Light	Control Panel → Sensor Calibration
Clear	Control Panel → Sensor Calibration
Clear All Calibrations	Control Panel → Estimate Calibration
Clear Grid	Virtual Proof
Clipping	Virtual Proof → Selected Sample
CONTRAST CALCULATOR	Contrast
CONTRAST GRADE	Contrast
CONTRAST RANGE	Contrast, Split Grade
Contrast (slider)	Contrast, Split Grade → Tunables
Control Panel	Main tab
DATA MANAGER	Control Panel
Default	Control Panel → Filter Values
Density	Virtual Proof → Selected Sample
Dev % / Dev EV	Virtual Proof → Enlarger mode
DODGE & BURN	Exposure, Contrast, Split Grade
Dodge and burn slider	Exposure, Contrast, Split Grade
Enlarger Mode	Virtual Proof → Settings
Equivalent grade	Split Grade
Estimate Calibration	Control Panel → Estimate Calibration
EV Range	Virtual Proof → Selected Sample
EXPOSURE CALCULATOR	Exposure
EXPOSURE TIME	Contrast

Exposure (slider)	Contrast, Split Grade → Tunables
Exposure Note	Contrast
Factor	Control Panel → Filter Values
Filter Brand	Sidebar
FILTER GRADE TRANSFORM	Exposure
Filter Grade	Contrast; Virtual Proof selector
FILTER VALUES	Control Panel
F-Stop	Exposure, Contrast, Split Grade
Fully charged	Sidebar → Battery
Grid Size	Virtual Proof
Grid width / height	Virtual Proof
HARD EXPOSURE	Split Grade
Help	Sidebar
Help Center	Help screen
HIGHLIGHTS	Contrast, Split Grade
HISTOGRAM	Virtual Proof
Intergration Time	Sidebar
ISO P	Control Panel → Filter Values
ISO R	Control Panel → Filter Values; Virtual Proof readout
ISO R Range	Control Panel → Paper & Filter Editor
Lengthening Factor	Control Panel → Paper & Filter Editor
Load Session	Control Panel → Data Manager
Load settings	Control Panel → Data Manager
Lux	Exposure; Virtual Proof
Match Quality	Contrast
Measure	Virtual Proof
Measure Highlight	Contrast, Split Grade
Measure LUX	Exposure
Measure Shadow	Contrast, Split Grade
Measure next (n/total)	Virtual Proof during a scan
Midpoint LUX	Contrast
Midpoint Time	Contrast
Midtones	Split Grade → Tunables
Min Delta (lux)	Virtual Proof → Settings
Min Stable Reads	Virtual Proof → Settings
PAPER & FILTER EDITOR	Control Panel
Paper Brand	Sidebar
Paper Type	Sidebar
Range / Mean / Min–Max	Virtual Proof → Enlarger mode
Reference Paper	Control Panel → Estimate Calibration
Reset	Contrast, Split Grade, Control Panel

SafelightIO v.1.0	Sidebar
Saved per paper	Contrast, Split Grade
Save Session	Control Panel → Data Manager
Save Settings	Control Panel → Data Manager
SELECT ZONE OF MEASUREMENT	Exposure
Select Filter	Control Panel → Paper & Filter Editor
SELECTED SAMPLE	Virtual Proof
Sensor Averaging Samples	Sidebar
Sensor Gain	Sidebar
Set Reference	Virtual Proof → Enlarger mode
Set V Zone	Virtual Proof
SHADOWS	Contrast, Split Grade
SOFT EXPOSURE	Split Grade
Stability Detection	Virtual Proof → Settings
Tolerance (%)	Virtual Proof → Settings
TOTAL EXPOSURE TIME	Split Grade
TOTAL TIME	Exposure, Contrast, Split Grade
TUNABLES	Contrast, Split Grade
Uniformity Limit	Virtual Proof → Settings
Verdict	Virtual Proof → Enlarger mode
Zone	Exposure; Virtual Proof readout

15.4 One-page cheat sheet

I want to…	Do this
Get a printing time	Exposure → select paper → zone V → **Measure LUX** → read time
Print a shadow at Zone III	Exposure → zone III → measure the shadow
Choose a grade	Contrast → measure highlight (dim) and shadow (bright) → read grade
Split-grade a contrasty negative	Split Grade → two readings → use soft + hard times
Preview the print	Virtual Proof → Grid Size → Measure each cell → read histogram
Check the enlarger	Virtual Proof → Enlarger Mode → 3×3 → measure → Set Reference
Calibrate a paper	Chapter 10: test strip, enter lux and time
Calibrate the sensor	Control Panel → Calibrate Dark, then Calibrate Light
Save my work	Control Panel → Save Session; Save Settings for a full backup
Update firmware	Press the BOOTSEL hole with a pin, connect USB-C, copy the `.uf2` to **RP2350**